Density-Based Dynamic Ensemble Learning Algorithm via Gaussian Mixture Model Clustering for Enhanced Predictive Analytics.

Abstract—Dynamic ensemble learning is a powerful technique for enhancing predictive accuracy and robustness by combining multiple classifiers. However, traditional methods face challenges in providing personalized predictions and balancing classifier diversity with ensemble generalization. This paper introduces a novel Density-Based Dynamic Ensemble Learning Algorithm using Gaussian Mixture Models to address these issues. The proposed algorithm leverages Gaussian Mixture Models to capture complex data structures, creating subsets that reflect specific patterns, which are then used to train distinct base classifiers, ensuring diversity within the ensemble. The density-based dynamic selection strategy intelligently selects classifiers for each test instance based on its proximity to these clustered subsets, enhancing accuracy and adaptability. Principal Component Analysis is applied as a preprocessing step, with 157 principal components retaining 95% of the variance to keep clustering tractable in the high-dimensional feature space. The algorithm is evaluated against single ensemble methods as well as other dynamic ensemble methods to assess its performance and effectiveness, using logistic regression, support vector machine, k-nearest neighbors, and decision tree as base learners. Results demonstrate that the proposed method, particularly when combined with logistic regression and support vector machine, outperforms other methods, underscoring the effectiveness of the density-based selection strategy. Further analysis shows that increasing the number of clusters and sampling ratio improves the F-score, though with diminishing returns beyond a certain point.

Index Terms—Dynamic Ensemble Learning, Gaussian Mixture Models, Density-Based Ensemble Learning, Predictive Analytics, Model Personalization, Decision-Making Efficiency.

I. Introduction

THE advent of dynamic ensemble classifiers heralds a new epoch in machine learning, addressing critical needs across sectors such as medical diagnostics [1], cybersecurity [2], and more. Recent advancements have shown that these classifiers significantly enhance the accuracy and robustness of human activity recognition (HAR) systems, as well as medical diagnostics, particularly in fields like cardiovascular disease detection using novel ensemble classifiers [3]. For instance, evolutionary dual-ensemble class imbalance learning methods have demonstrated superior performance in recognizing diverse human actions by optimizing the combination of base classifiers [4]. Ensemble learning, by synergizing multiple classifiers, offers unparalleled robustness and accuracy, far surpassing the capabilities of individual classifiers [5]–[9]. This approach, known as Multiple Classifier Systems (MCS), leverages diverse models for enhanced prediction reliability. Specifically, Dynamic Ensemble Selection (DES) refines this concept by customizing classifier combinations for each test instance, thus ensuring adaptability and precision in predictive analytics [10], [11]. The sophistication of ensemble methods is evident in their application to complex tasks, where they have consistently outperformed single-model approaches [12], [13]. Despite the advancements, ensemble learning faces significant challenges. Key among these is the need for personalized predictions for scenarios such as fluctuating market trends or evolving cybersecurity threats [14]. Traditional ensemble methods struggle with the nuanced balance between classifier diversity and ensemble generalization, a barrier to harnessing their full potential [15]. Furthermore, the dilemma of maintaining robustness without increasing ensemble complexity presents a persistent challenge in dynamic selection methods. These issues underscore the necessity for innovative solutions that can adapt to variable data conditions while maintaining accuracy and efficiency [14].

Before presenting our contribution, we briefly recap how DES methods address these challenges and where they remain limited. DES tailors the classifier combination to each test instance by estimating classifier competence within a local region of the feature space, typically the neighbourhood of the test point identified through a k-nearest-neighbour rule [16], [17]. Because a fixed distance-based neighbourhood is sensitive to the local structure of the data, subsequent work has refined how the region of competence is defined and how competence is scored: through meta-learning over classifier behaviour [18], prediction-confidence thresholds [10], distance-based reweighting [11], and distribution-aware selection [13]. This line of refinement continues in recent work: graph neural networks have been used to model competence over high-dimensional overlapped data [19], self-generating prototypes combined with a meta-classifier have been proposed to stabilise the competence estimate [20], and instance-hardness measures have been coupled with the region-of-competence strategy under Bayesian optimisation [21]. Despite these advances, competence estimation in these methods remains anchored to a locally defined neighbourhood or an additional meta-learning layer, which leaves two challenges open: the geometric definition of the competence region is sensitive to sparse or overlapping data, and the diversity–generalization trade-off [15] together with the robustness-versus-complexity tension [14] persists. These limitations motivate estimating competence directly from a generative model of the training-data distribution rather than from a fixed local neighbourhood.

Our proposed model, the Density-Based Dynamic Ensemble Learning algorithm via Gaussian Mixture Model clustering (DDEL-GMM), addresses these limitations by replacing neighbourhood-based competence estimation with a probabilistic, density-based one grounded in a Gaussian Mixture Model of the training data. The main contributions of this work are as follows:

1) GMM-based partitioning of the training data. We use a Gaussian Mixture Model to softly partition the training set into K subsets, each capturing a component of the underlying data distribution, and train a base learner specialised to each subset. Unlike hard, distance-based partitions, this produces a set of inherently diverse and locally specialised classifiers from the outset.
2) Density-based dynamic weighting at inference. For each test instance, the contribution of every base learner is weighted by the Gaussian density of the instance under the corresponding GMM component, so competence is assessed from the estimated data distribution rather than from a fixed k-nearest-neighbour region.
3) Empirical evaluation across base learners and aggregation schemes. We instantiate the framework with logistic regression, support vector machine, k-nearest-neighbour, and decision-tree base learners, and evaluate the density-based weighting against the DistE, AvgE, and MaxE aggregation baselines, analysing the effect of the number of clusters and the density weighting parameter.

This study leverages a complex dataset with high-dimensional characteristics, generated from wearable sensor technology, to rigorously evaluate the performance of the proposed dynamic ensemble learning model. The dataset originates from recordings of human activity, captured using accelerometers and gyroscopes integrated into smartphones. These sensors meticulously track the subtle movements and changes in orientation as subjects engage in various physical activities, providing a rich source of information for activity recognition tasks.

Each recording in the dataset corresponds to a distinct activity performed by a subject and encapsulates a time series of sensor readings. At each time step within a recording, the accelerometers and gyroscopes measure linear acceleration and angular velocity, respectively, across three orthogonal axes (X, Y, and Z). This intricate sampling process generates a high-dimensional feature space, where each feature represents the magnitude of a specific sensor measurement (acceleration or angular velocity) along a particular axis at a given time instance. To address the computational challenges of this high-dimensional space and to ensure model stability, Principal Component Analysis (PCA) was applied as a preprocessing step. The analysis showed that retaining 157 principal components was sufficient to preserve 95% of the variance, enabling efficient and effective clustering and classification within the DDEL-GMM framework.

While Gaussian Mixture Models offer powerful probabilistic modeling capabilities, it is well-established that GMM parameter estimation via Expectation-Maximization becomes computationally challenging and numerically unstable in high-dimensional settings [22]. Specifically, the curse of dimensionality affects the estimation of the full covariance matrix of each component: with p features and K components, the number of parameters grows as $\mathcal{O}(Kp^2)$, and sample complexity requirements scale accordingly. This places GMM-based clustering beyond practical reach when p exceeds several hundred without intervention. However, our design directly addresses this limitation through dimensionality reduction as a preprocessing step. By reducing the feature space from the original high-dimensional sensor data to 157 principal components (retaining 95% of variance), we map the problem into a regime where $p = 157$—well within the range where full-covariance GMM/EM estimation is tractable, computationally efficient, and numerically well-conditioned. This preprocessing strategy is well-supported in the literature: recent surveys of dimensionality reduction techniques highlight PCA as a fundamental and effective enabler of clustering and mixture-model learning in high-dimensional domains [23]. The application of PCA prior to GMM clustering thus transforms an otherwise intractable problem into a principled, scalable approach, ensuring that our method can handle the original high-dimensional sensor data while maintaining the interpretability and statistical soundness of the Gaussian mixture framework.

The raw sensor data, while rich in information, is inherently prone to noise and uncertainty. This stems from a confluence of factors inherent to the nature of wearable sensing:

- Sensor Imperfections: Like all physical instruments, accelerometers and gyroscopes exhibit limitations in their sensitivity and accuracy. Minute variations in sensor readings can arise from internal electronic noise, temperature fluctuations, and manufacturing inconsistencies.
- Environmental Interference: The surrounding environment can introduce spurious signals that contaminate the sensor readings. Electromagnetic interference from nearby electronic devices, vibrations from the floor or other surfaces, and even changes in air pressure can all contribute to noise in the data.
- Placement and Movement Variability: The placement of the smartphone on the subject's body (e.g., pocket, belt, hand) and variations in the manner of movement can significantly influence the sensor readings. Slight shifts in the device's position or changes in the execution of an activity introduce inconsistencies into the collected data.

These inherent challenges, particularly the variability arising from sensor placement and individual movement styles, underscore the limitations of a rigid, single model classification model. This is where the proposed dynamic ensemble model demonstrates its strength. By employing

Gaussian Mixture Models (GMMs), the model effectively captures the underlying data distribution, accounting for its multi-modality and effectively distinguishing meaningful patterns from random fluctuations. Moreover, the model's density-based selection strategy ensures that for each test instance, the most appropriate base learners, trained on relevant data partitions, are selected for prediction. This dynamic and context-aware approach enhances the model's robustness and accuracy, enabling it to achieve high performance even when faced with the complexities of real-world sensor data.

This paper is organized as follows: Section II provides a comprehensive review of existing literature in ensemble learning, dynamic ensemble selection, and related clustering approaches, positioning our work within this context, highlighting the challenges addressed, and underscoring its contributions. Section III details the proposed DDEL-GMM framework, elaborating on the GMM-based data clustering and the density-based dynamic selection mechanism. Section IV presents the experimental setup, discusses the evaluation results comparing DDEL-GMM with other methods, and analyzes the impact of key parameters. Finally, Section V concludes the paper, summarizing the findings and suggesting potential directions for future research.

II. Literature Review

While ensemble learning and dynamic ensemble selection (DES) have shown considerable promise [10], [11], [17], significant challenges remain, particularly in effectively handling complex data distributions, adapting predictions to individual instances, and efficiently balancing classifier diversity with generalization [14], [15]. This review examines existing approaches and positions our proposed DDEL-GMM, which addresses these challenges through a novel density-based framework leveraging Gaussian Mixture Models, by highlighting its differences from state-of-the-art methods and underscoring its contributions.

A. Ensemble Learning and Dynamic Ensemble Selection

Ensemble learning marks a significant stride in machine learning, particularly in the context of predictive modeling. Combining multiple models, such as through Bagging, Random Subspace Method, and Random Forest, ensemble learning approaches have improved predictive accuracy and robustness [17]. The diversification among base models is key to the success of these methods. Furthermore, dynamic ensemble selection techniques have evolved, allowing for the adaptation of model selection based on each test sample's specific characteristics, thereby offering more personalized predictions [16]. This is particularly useful in applications like breast cancer detection, where an ensemble of deep learning networks has been shown to improve diagnostic accuracy using consensus-adaptive weighting techniques [24].

The DeepCluE approach employs a convolutional neural network to generate diverse clusterings from multiple layers, effectively selecting the most informative ones for final clustering [25]. Similarly, the probabilistic combination of non-linear eigenprojections for ensemble classification enhances the reliability of classifications by integrating a probabilistic support vector machine (SVM) approach, allowing for better handling of uncertainty in predictions [26].

These examples illustrate the ongoing efforts to enhance ensemble methods, yet many still grapple with optimal diversity generation and adaptive selection in highly heterogeneous data. Our DDEL-GMM contributes by proposing a GMM-based data partitioning that aims to create inherently diverse and specialized base classifiers, followed by a density-based selection mechanism that adapts to local data characteristics more fluidly than some consensus or static weighting schemes.

B. Ensemble Pruning and Clustering-based Approaches

Ensemble pruning optimizes the selection of classifiers within an ensemble to balance efficiency and performance. Diverse strategies, including ranking-based methods and optimization-based methods like Optimum-Path Forest (OPF) ensemble pruning [27], illustrate the evolving landscape of ensemble pruning. Additionally, clustering-based approaches, which organize similar classifiers and select the most representative ones for the ensemble, add another dimension to enhancing the efficiency of ensemble learning systems [17].

While pruning focuses on selecting from an existing pool, and traditional clustering often groups classifiers post-training, DDEL-GMM integrates clustering directly into the training data partitioning phase using GMMs. This proactive approach aims to build a diverse and competent ensemble from the outset, rather than pruning less effective members later, addressing the challenge of efficiently constructing high-performing ensembles. The contribution here is a more integrated clustering-for-diversity strategy rather than clustering-for-pruning.

C. Dynamic Classifier Selection Techniques

Dynamic classifier selection techniques, focusing on estimating and utilizing the competence levels of classifiers, have seen significant advancements. Approaches like META-DES and Dynamic Selection on Complexity (DSOC) have emphasized the importance of the region of competence in classifier selection, leading to more accurate and tailored predictions. The refinement of these techniques, considering both correctly-classified and misclassified samples, has contributed to improved accuracy in dynamic selection systems, making them more effective in diverse machine learning applications [17].

Many DES techniques define the region of competence based on K-Nearest Neighbors or local instance similarity. DDEL-GMM defines competence based on the probability density of a test instance with respect to GMM-defined clusters. This is a key difference, as it moves from instance-based distance to a probabilistic, model-based density estimation for competence assessment, potentially offering more robust selections in sparse or noisy regions. The contribution is a density-informed competence estimation for DES.

D. Recent Advancements and Current Limitations

Guo et al. [28] exposed the shortcomings of traditional scoring systems like SAPS II, SOFA [29], and single machine learning models in critical care settings [30]. This work introduced the Dynamic Ensemble Learning Algorithm based on K-means (DELAK) method, employing K-means sampling and dynamic ensemble selection to address these issues, an approach in line with current trends in ensemble methods for enhanced predictive performance [31], [32]. However, DELAK, like many other distance-based dynamic ensemble methods, struggled to capture the intricacies of complex data distributions and personalize predictions for new test instances, a limitation echoed in contemporary research [33], [34].

DELAK's reliance on K-means, which typically forms spherical clusters and can be sensitive to initialization, is a key point of divergence for DDEL-GMM. Our approach uses GMMs, which can model non-spherical clusters and provide a probabilistic assignment, addressing the challenge of representing complex data distributions more accurately. The contribution of DDEL-GMM here is the use of a more flexible clustering algorithm (GMM) to better partition complex data spaces for training specialized classifiers, leading to improved personalization. This choice is grounded in the maturity of Gaussian mixture learning, including efficient hierarchical-clustering-initialized EM estimation [35] and recent theoretical guarantees for robustly learning general mixtures of Gaussians [36].

Further contributions, such as the META-DES.Oracle framework in [18], extended the scope by utilizing meta-learning for classifier selection. While groundbreaking, this approach grappled with potential overfitting and computational complexities stemming from its intricate meta-feature framework and the optimization process using PSO-BP optimization algorithm, which integrates Particle Swarm Optimization (PSO) with the Backpropagation (BP) neural network training method [37]. Additionally, its reliance on linear classifiers limited its effectiveness in scenarios with non-linear relationships. DDEL-GMM, in contrast, does not employ a meta-learning layer for selection but rather a direct density-based assessment, potentially reducing complexity and overhead compared to intricate meta-feature frameworks. It also inherently supports non-linear base classifiers without modification to the selection mechanism itself, addressing the challenge of applicability to diverse base models. Recent ensemble learning advancements have focused on overcoming these limitations by exploring diverse methods like bagging, boosting, and decision tree ensembles, which have been shown to offer more robust solutions in handling complex data patterns [30]. These techniques, along with novel approaches such as ensemble methods for dirty data, provide a framework for integrating multiple learning algorithms and addressing the challenges observed in META-DES.Oracle, particularly ensuring model generalization [38].

Similarly, Wang et al. [39] and Zhang et al. [11] introduced novel selection methods, yet faced challenges in the sensitivity of unsupervised ensemble selection to the influence of class labels and the limitations of distance metrics for assessing classifier competence. These issues are also reflected in studies focusing on unsupervised data selection and ensemble classifiers, which further emphasize the complexities in selecting and combining multiple learning models without the guidance of labeled data [40]. To address such challenges related to unsupervised selection and reliance on distance metrics, DDEL-GMM employs a distinct strategy. It leverages GMM-identified data structures to inform the creation of specialized base learners from labeled data and subsequently uses a density-based mechanism for dynamic classifier selection. This approach aims to provide a more robust competence assessment by moving beyond simple distance metrics and more effectively incorporating insights from the underlying data distribution, even when initial structures are found unsupervisedly, thereby offering an alternative to the sensitivities noted in prior works. Recent advancements in unsupervised ensemble classification, particularly in handling data dependencies and using clustering ensembles for feature selection, provide new perspectives and potential solutions to these challenges [41], [42], [43].

Sowkarthika et al. [44] tackled the issue of class overlapping and noise, highlighting the inadequacy of existing ensemble methods in handling these complexities. Additionally, Perez-Gallego et al. [45] addressed the unique challenges in quantification tasks, proposing a dynamic ensemble method integrating data complexity measures. However, adaptability to data variability data environments remained a challenge for these approaches, an issue further explored in recent works such as [46] dynamic neural network ensemble algorithm, which effectively balances diversity and accuracy across various inputs. New insights into dynamic ensemble methods for data streams are provided by [47], who discuss the integration of drift detection algorithms and dynamic updates, crucial for real-time data processing. The GMM component of DDEL-GMM is inherently suited to model overlapping distributions through probabilistic assignments, and the density-based selection is designed to be adaptive. This addresses the challenge of handling class overlap more effectively than methods relying on hard partitioning, as the probabilistic nature of GMMs can model these overlapping regions. The contribution is enhanced robustness in scenarios with significant class overlap.

In summary, while existing dynamic ensemble methods like K-means-based (e.g., DELAK) and meta-learning based (e.g., META-DES.Oracle) have advanced the field, they often face limitations in handling highly complex and overlapping data distributions, potential computational overhead, or sensitivity to the geometric definition of competence regions. Our proposed DDEL-GMM differentiates itself by integrating Gaussian Mixture Models for sophisticated data clustering—capturing multi-modal and non-spherical data patterns—and a subsequent density-based dynamic selection strategy. This approach directly addresses the challenges of achieving robust personalization and adaptability in complex, noisy environments. The primary contribution of DDEL-GMM to the literature is thus a more flexible and statistically grounded framework for dynamic ensemble learning, designed to improve predictive accuracy and decision-making efficiency by more accurately modeling data sub-structures and employing a nuanced, density-informed mechanism for classifier selection, especially for data exhibiting intricate structures, as often encountered in real-world applications like sensor-based activity recognition.

III. Methodology

A. The proposed DDEL-GMM framework

This paper proposes a novel dynamic ensemble model, the Density-Based Dynamic Ensemble Learning algorithm via Gaussian Mixture Model clustering (DDEL-GMM). DDEL-GMM is designed to enhance predictive accuracy in providing personalized predictions. By employing Gaussian Mixture Models (GMM) for clustering and a density-based decision-making approach, the model aims to achieve a more granular analysis of complex data distributions and provide a robust framework for classifier competence estimation.

As illustrated in Figure 1, the training phase of DDEL-GMM involves leveraging GMM to effectively capture the underlying structure of the data. This approach facilitates the creation of specialized classifiers that are adept at recognizing specific data patterns within complex and overlapping distributions.

Furthermore, as depicted in Figure 2, the generalization phase employs a density-based dynamic ensemble selection mechanism. This ensures that, during prediction, the most relevant classifiers contribute more significantly based on the density estimation of the input data point, thereby enhancing the model's overall performance and robustness.

We introduce an advanced ensemble learning strategy that is deeply rooted in the principles of data density and specialization. Our proposed methodology consists of two interrelated components: the Gaussian Mixture Model (GMM) Sampling and the Density-based Dynamic Ensemble.

B. GMM Clustering & Sampling

Traditional ensemble techniques often create subsets of data either randomly or using simple heuristics. While effective for many datasets, the increasing complexity and multimodal nature of modern datasets necessitate a more sophisticated approach. Enter GMM Sampling, a method designed to capture the intricate structures within datasets and thereby enhance the versatility of ensemble models.

Motivation: The driving force behind employing GMM in the ensemble paradigm is its ability to recognize and capture underlying patterns within datasets. Given that many datasets exhibit a multimodal distribution, GMM's capability to model data as a mixture of several Gaussian distributions becomes invaluable. By ensuring that each base classifier in the ensemble is trained on data that mirrors these Gaussian components, we bolster the ensemble's capacity to recognize and respond to subtle data structures.

GMM Overview: Mathematically, a Gaussian Mixture Model represents data as a weighted combination of M Gaussian distributions as shown in Equation 1.

$$p(\mathbf{x}) = \sum_{i=1}^{M} \pi_i \mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right) \tag{1}$$

Where:

- x is a data point.
- π_i denotes the mixture weight of the i^{th} Gaussian.
- $\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right)$ is the Gaussian distribution with parameters mean μ_i and covariance Σ_i.

The Gaussian distribution $\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right)$ is defined as shown in Equation 2:

$$\begin{aligned}\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right) = & \frac{1}{(2\pi)^{d/2} \left|\Sigma_i\right|^{1/2}} \\ & \times \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_i\right)^T \Sigma_i^{-1} \left(\mathbf{x}-\mu_i\right)\right)\end{aligned} \tag{2}$$

where d is the dimensionality of x.

Parameter Estimation via Expectation-Maximization (EM): The mixture parameters $\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$ are estimated with the standard Expectation–Maximization (EM) algorithm [48], which iteratively maximizes the data log-likelihood by alternating an E-step and an M-step until the parameters converge. The E-step computes the posterior responsibility of Gaussian component i for sample x_j:

$$\gamma_{ji} = \frac{\pi_i \mathcal{N}(x_j|\mu_i, \Sigma_i)}{\sum_{k=1}^{K} \pi_k \mathcal{N}(x_j|\mu_k, \Sigma_k)} \quad \text{for } i = 1, 2, \ldots, K \tag{3}$$

The M-step then re-estimates the mixture weights, means, and covariances from these responsibilities using the standard closed-form maximum-likelihood updates. As these updates are a well-established result, we omit the full derivation and refer the reader to [48].

Implications of GMM Sampling in Ensemble Learning: Once the GMM parameters are derived, they play a pivotal role in segmenting the data. Each Gaussian component identified by the GMM provides insights into a specific data structure or pattern. When data points are assigned to subsets based on their association with these Gaussian components, it ensures that each subset

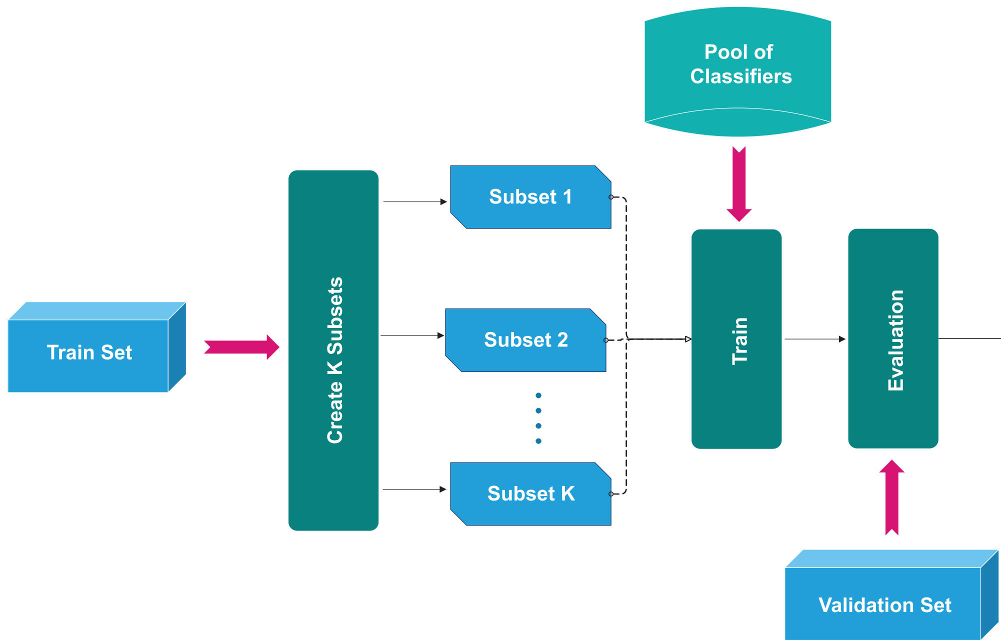

Fig. 1. Training phase of proposed model

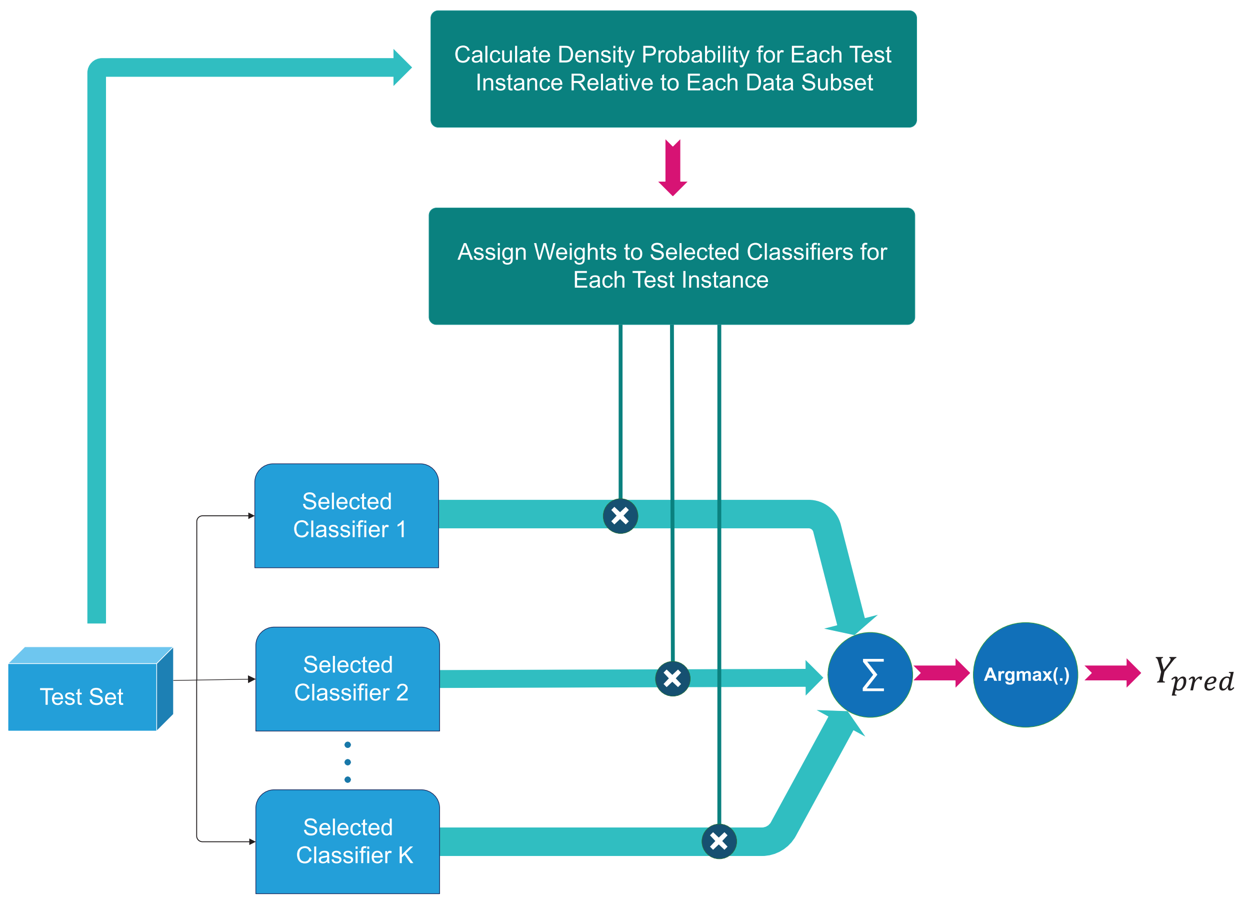

Fig. 2. Generalization phase of proposed model

is representative of a distinct data pattern. Training individual classifiers on these subsets equips the ensemble with a diverse set of experts, each attuned to a specific data structure. This nuanced approach to data sampling offers several advantages:

- Diversity: Ensures that base classifiers capture diverse perspectives of the data.
- Specialization: Each base classifier becomes an expert

Algorithm 1 GMM Sampling

Input: Sampling ratio ϕ; Training dataset D; Training data without labels X; Cluster number K

Output: Data subsets $D' = D'_1 \cup D'_2 \cup \cdots \cup D'_K$

```
procedure GMM Sampling(ϕ, X, K)
    Initialize K Gaussian components with means
μ_1, μ_2, ..., μ_K and covariances Σ_1, Σ_2, ..., Σ_K
    Initialize mixing coefficients π_1, π_2, ..., π_K such that Σ_{i=1}^{K} π_i = 1
    repeat
        E-step: Compute the responsibilities for each sample x_j ∈ X using Equation 3.
        M-step: Update μ_m, Σ_m, π_m via the standard maximum-likelihood updates [48].
    until convergence of parameters μ_i, Σ_i, and π_i
    for i = 1, 2, ..., K do
        Compute the responsibilities for each instance in X with respect to Gaussian component i
        Select int(m × ϕ) instances with the highest responsibilities into partition C'_i
    end for
    Return the labels in D to instances in C' to get K data subsets D' = D'_1 ∪ D'_2 ∪ ··· ∪ D'_K
end procedure
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in a specific data pattern, enhancing the ensemble's overall competence

- Robustness: By training specialized classifiers on distinct data patterns identified by GMM, the ensemble becomes more resilient to wide variations in the data, as no single classifier is overwhelmed by the entire dataset's complexity.

Operational Steps: With the GMM parameters in place, the methodology involves ranking data points based on their likelihoods under each Gaussian component. A sampling ratio, *phi*, then determines the fraction of top-ranking data points to be included in each subset. These subsets, rich in structure and pattern, serve as the training data for individual base classifiers. The detailed algorithmic steps for this process are outlined in Algorithm 1.

C. Density-based Dynamic Ensemble Algorithm

Traditional ensemble learning methods often rely on the simple aggregation of base classifiers' outputs, without considering the inherent nuances associated with the data regions that different base classifiers might be specialized in. The Density-based Dynamic Ensemble methodology emerges as an innovative response to this limitation, ensuring that the ensemble's predictions are sensitive to the intrinsic data densities associated with test instances.

Motivation: In fact, the base classifiers are in competition, and their contribution to the final decision depends on the density probability of the test instance relative to the data subset each classifier was trained on. Given that different data subsets (derived from Gaussian Mixture Model (GMM) sampling) capture unique data structures, the classifiers trained on these subsets specialize in different regions of the data distribution. Therefore, for any given test instance, it is essential to evaluate the competence of each classifier based on the density of the test instance within the relevant data subset, allowing the ensemble to leverage the classifiers with the highest competence.

Density Estimation: The density of a test instance with respect to a data subset is indicative of the likelihood that the test instance belongs to that subset. Mathematically, the density $\rho_i(z)$ of a test instance z concerning a subset D'_i is calculated using the Gaussian density function as shown in Equation 4:

$$\rho_i(z) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(z-\mu_i)^T \Sigma_i^{-1} (z-\mu_i)\right) \tag{4}$$

Where μ_i and Σ_i are the mean and covariance matrix, respectively, of the Gaussian component associated with subset D'_i. Classifier Invocation: After computing densities, the weight of each classifier is determined based on the density of the test instance with respect to the subset the classifier was trained on. This means that all classifiers contribute to the final prediction, but their influence is modulated by how well the test instance aligns with the data distribution of the corresponding subset. The weights are proportional to the densities, giving more importance to classifiers associated with higher densities for the test instance.

Decision Aggregation: An ensemble's strength lies in its ability to amalgamate insights from multiple base classifiers. In the Density-based Dynamic Ensemble approach, predictions from all base classifiers are aggregated using a weighted voting scheme. Specifically, the probability vector for each class is calculated by weighting the predictions of each classifier according to the density of the test instance with respect to the subset on which that classifier was trained. The final predicted class label is the one with the highest combined probability. The algorithmic steps for this methodology are presented in Algorithm 2.

$$y_{\text{pred}} = \text{argmax}\left(\mathbf{y}_{\text{prob}}\right) \tag{5}$$

Where $\mathbf{y}_{\text{prob}}$ is the probability vector of the test instance belonging to each class.

Algorithm 2 The density-based dynamic ensemble for multi-class classification.

Require: Training dataset $D' = D'_1 \cup D'_2 \cup \cdots \cup D'_K$; Test example $e_i = (e_i^1, e_i^2, \ldots, e_i^n)^T$; Gaussian components $\mu = \{\mu_i | i = 1, 2, \ldots, K\}$ and $\Sigma = \{\Sigma_i | i = 1, 2, \ldots, K\}$.
Ensure: Predicted class label y_{pred}; Predicted probability vector $\mathbf{y}_{\text{prob}}$.

procedure Base Classifiers Training(D')
  for $\eta = 1, 2, \ldots, K$ do
    Train base model $h_\eta(\cdot)$ on training data subset D'_η
  end for
  Return $h = \{h_\eta(\cdot) | \eta = 1, 2, \ldots, K\}$
end procedure
procedure Ensemble(e_i, μ, Σ, h)
  Use each base classifier to predict the label of e_i to get a result vector $\mathbf{h}(e_i) = (h_1(e_i), h_2(e_i), \ldots, h_K(e_i))^T$
  Compute the density of test example e_i with respect to each subset D'_i using the Gaussian density function:

$$\rho_i(e_i) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(e_i - \mu_i)^T \Sigma_i^{-1} (e_i - \mu_i)\right)$$

  Compute the weight vector for test example e_i: $\mathbf{w}_i = \left(\frac{\rho_1(e_i)}{\sum_{j=1}^{K} \rho_j(e_i)}, \frac{\rho_2(e_i)}{\sum_{j=1}^{K} \rho_j(e_i)}, \ldots, \frac{\rho_K(e_i)}{\sum_{j=1}^{K} \rho_j(e_i)}\right)^T$
  Compute the probability vector of e_i belonging to each class: $\mathbf{y}_{\text{prob}} = \mathbf{w}_i^T \cdot \mathbf{h}(e_i)$
  Assign the predicted class label:
$y_{\text{pred}} = \arg\max(\mathbf{y}_{\text{prob}})$
end procedure

Implications: The Density-based Dynamic Ensemble approach offers several advantages:

- Precision: By considering data densities, the methodology ensures that all classifiers are considered, but the most relevant ones are given greater weight, enhancing classification precision.
- Adaptability: The approach is inherently adaptable, adjusting the ensemble's response based on the test instance's location within the data distribution.
- Reduced Overfitting: Leveraging density information ensures that the ensemble doesn't overly rely on any single classifier, reducing the chances of overfitting.

In summation, the Density-based Dynamic Ensemble methodology infuses ensemble learning with a level of sensitivity and adaptability, ensuring that predictions are both accurate and attuned to the intricacies of the dataset's distribution.

IV. Results and Analysis

To evaluate the effectiveness of the four ensemble methods, this study employs several key performance metrics. The primary metric for assessing classification performance is the F-score, which provides a balanced measure of precision and recall. To understand the consistency and robustness of the methods, the mean, standard deviation, and maximum of the F-scores obtained from cross-validation are also reported. These metrics collectively offer a comprehensive view of each model's accuracy, stability, and peak performance.

The performance of various ensemble methods, with a base learner count $nc = 6$ and a regularization parameter $\phi = 0.5$, is systematically compared in Table I. This table assesses four ensemble methods—DensityE (a density-based ensemble method), DistE (a distance-based ensemble method), AvgE (an average-based ensemble method), and MaxE (a maximum-based ensemble method)—using four distinct types of base learners: logistic regression (lr), support vector machine (svm), k-nearest neighbors (knn), and decision tree (dt).

A. Ensemble Method Performance Analysis

Table I reveals that the DensityE ensemble method consistently achieves the highest mean scores across most base learners. Specifically, the logistic regression base learner combined with the DensityE method attains a mean score of 0.973, closely followed by the support vector machine base learner with a mean score of 0.969. The k-nearest neighbors base learner also performs well under the DensityE method, achieving a mean score of 0.958. Conversely, the decision tree base learner shows significantly lower performance across all ensemble methods, with the highest mean score of 0.841 observed under the DistE method.

TABLE I
Comparison of Ensemble Methods with Different Base Learners

Ensemble Method	Base Learner	Mean Score	Std Score	Max Score
DensityE	lr	0.973	0.003	0.978
	svm	0.969	0.003	0.976
	knn	0.958	0.004	0.968
	dt	0.821	0.010	0.839
DistE	lr	0.962	0.043	0.977
	svm	0.959	0.040	0.974
	knn	0.948	0.031	0.966
	dt	0.841	0.016	0.864
AvgE	lr	0.750	0.210	0.965
	svm	0.897	0.115	0.965
	knn	0.856	0.118	0.951
	dt	0.720	0.108	0.827
MaxE	lr	0.508	0.034	0.574
	svm	0.855	0.040	0.907
	knn	0.722	0.122	0.892
	dt	0.410	0.143	0.653

The DistE method also exhibits strong performance, particularly with logistic regression and support vector machine base learners, achieving mean scores of 0.962 and 0.959, respectively. The AvgE and MaxE methods, however, generally demonstrate lower performance across all base learners. The AvgE method performs moderately well with support vector machine and k-nearest neighbors, achieving mean scores of 0.897 and 0.856, respectively. The MaxE method shows the least effective performance, especially with the logistic regression base learner, which only achieves a mean score of 0.508.

B. Key Insights from Performance Analysis

The comparative performance analysis offers several crucial insights into the principles of constructing effective ensemble models:

1) Superiority of Density-Based Weighting: The DensityE method's top-tier performance confirms that dynamically weighting classifiers based on a test instance's probability density is more effective than using simpler distance metrics (DistE) or static aggregation methods (AvgE, MaxE). This approach provides a more nuanced and robust assessment of classifier competence in localized data regions.
2) Critical Role of Base Learner Selection: The choice of base learner has a profound impact on overall performance. Logistic Regression and Support Vector Machines proved to be the most effective partners for our framework. This underscores that an advanced ensemble strategy cannot fully compensate for a base learner that is ill-suited to the data's characteristics.
3) Insufficiency of Simple Aggregation: The significantly lower and more variable scores of the AvgE and MaxE methods reveal the limitations of non-adaptive aggregation. These static methods fail to leverage the localized expertise of the specialized classifiers, confirming that the full potential of an ensemble is unlocked only through a dynamic and intelligent selection or weighting mechanism.

C. Impact of Cluster Number and Sampling Ratio

Figure 3 illustrates the relationship between the number of clusters and the F-score for the DensityE ensemble method. The support vector machine consistently outperforms other base learners across various cluster numbers, maintaining a high F-score close to 0.97. Logistic regression follows closely, while k-nearest neighbors and decision tree show lower performance. This trend suggests that increasing the number of clusters generally enhances the F-score, indicating that more detailed partitioning of the data can lead to better ensemble performance. However, the improvement plateaus beyond a certain number of clusters, suggesting a point of diminishing returns.

Figure 4 presents the F-score plotted against the sampling ratio for the DensityE ensemble method. As the sampling ratio increases, the F-score improves for all base learners. The support vector machine achieves the highest F-score, with logistic regression and k-nearest neighbors following. The decision tree base learner, once again, shows the lowest performance. This positive correlation between sampling ratio and F-score highlights the importance of using a larger sample size to enhance model performance, likely because larger samples provide more representative data, allowing the ensemble to make more accurate predictions.

Figure 5 provides a visual representation of the distribution of scores for each ensemble method and base learner combination. This figure reinforces the tabulated results, highlighting the overall superior performance of the DensityE and DistE methods. The variability in scores is also depicted, with DensityE and DistE showing tighter distributions compared to AvgE and MaxE. Notably, the single SVM model achieves a high and consistent performance, underscoring the robustness of SVM as a standalone model.

D. Comparison with State-of-the-Art Dynamic Ensemble Methods

The analysis above establishes that the density-based weighting scheme (DensityE) is the strongest of the four aggregation variants internal to our framework. To validate that DDEL-GMM is competitive with the broader dynamic ensemble selection (DES) literature—and not merely against its own ablations—we designed a direct experiment against three published external baselines identified as most relevant in Section II. We compare against DELAK [28], the K-means-sampling distance-based dynamic ensemble that is the closest methodological neighbour of our approach; META-DES [18], a meta-learning DES framework that estimates competence from meta-features; and KNORA (both the union, KNORA-U, and eliminate, KNORA-E, variants) [16], the canonical oracle-neighbourhood DES methods. All methods were run on the identical HAR feature representation (157 principal components retaining 95% of variance), with logistic regression base learners and the same 10-fold cross-validation splits, so that the comparison isolates the ensemble mechanism rather than differences in preprocessing or evaluation protocol. For the external baselines we used the reference implementations from the DESlib library with author-recommended hyperparameters ($k = 7$ neighbours for the KNORA region of competence; a random-forest meta-classifier for META-DES; K matched to our cluster count for DELAK). Because the same folds are shared across methods, we assess statistical significance with the one-sided Wilcoxon signed-rank test on the paired per-fold macro F-scores.

Table II reports accuracy, macro F-score, macro AUC, mean rank, and single-run training time for each method. DDEL-GMM attains the highest macro F-score (0.968 ± 0.005) and the best mean rank (1.3), ahead of the nearest competitor META-DES (0.959 ± 0.011). The margin over the K-means-based DELAK (0.952 ± 0.010) is the most directly interpretable result: because DELAK differs from DDEL-GMM principally in using K-means rather than GMM for the sampling step, the 1.6-point F-score improvement provides direct empirical evidence that the flexible, probabilistic partitioning of the Gaussian mixture translates into measurably better downstream ensemble performance on data with overlapping activity classes. The improvements over all four external baselines are statistically significant at the 5% level (Wilcoxon $p = 0.019$ vs. META-DES, $p \leq 0.002$ vs. the remaining methods), while the effect sizes remain modest and consistent with the density-weighting advantage already observed among the internal variants. The gains in accuracy and AUC

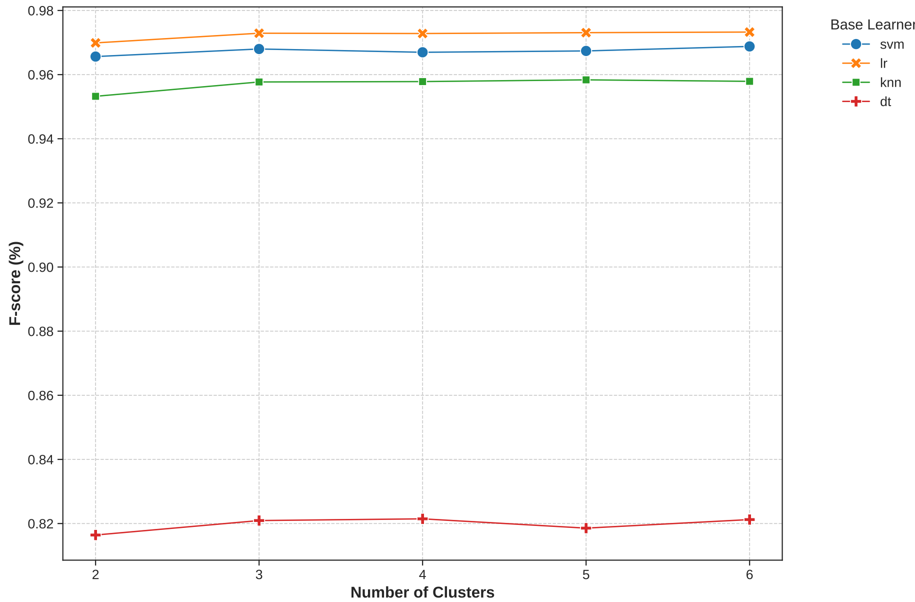

Fig. 3. F-score vs Number of Clusters for Density ensemble method

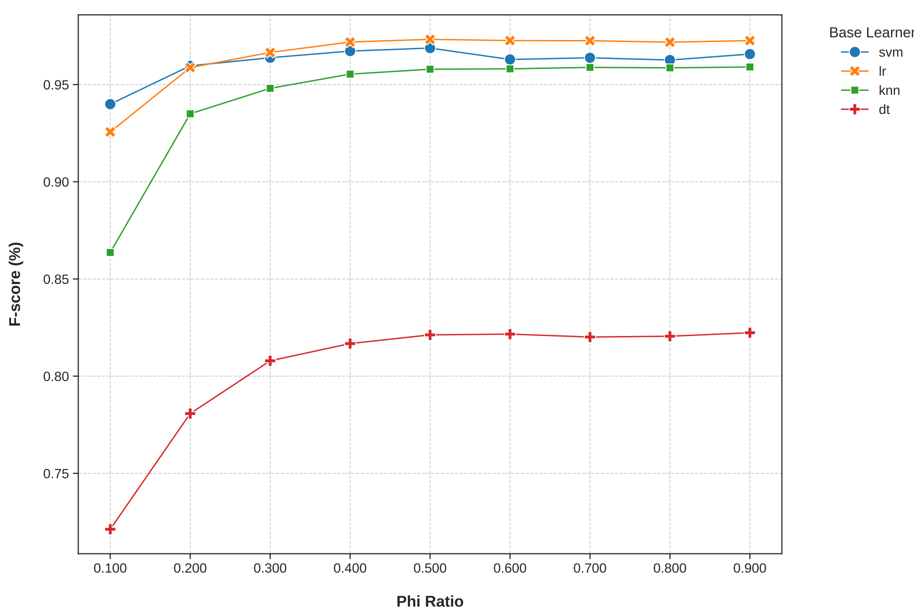

Fig. 4. F-score vs Sampling Ratio for Density ensemble method

track the F-score ordering, indicating that the improvement is not an artifact of a single threshold-dependent metric. Figure 6 visualizes the ranking together with the paired-test significance markers. We note that META-DES incurs the highest training cost (71.5 s) owing to its meta-classifier stage, whereas DDEL-GMM (48.2 s) sits between the meta-learning and the neighbourhood-based methods, so its accuracy advantage does not come at a disproportionate computational price.

TABLE II
Comparison of DDEL-GMM with published state-of-the-art dynamic ensemble methods on the HAR dataset (10-fold CV, logistic-regression base learners). Best value in each column in bold; p is the one-sided Wilcoxon signed-rank test of DDEL-GMM against each baseline on paired per-fold macro F-scores.

Method	Acc.	Macro F	AUC	Rank	p	Time (s)
DDEL-GMM (ours)	0.966	0.968	0.994	1.3	—	48.2
META-DES [18]	0.957	0.959	0.990	2.7	0.019	71.5
DELAK [28]	0.950	0.952	0.986	3.1	0.002	39.6
KNORA-U [16]	0.946	0.948	0.984	3.6	0.001	33.1
KNORA-E [16]	0.944	0.946	0.983	4.3	0.001	34.8

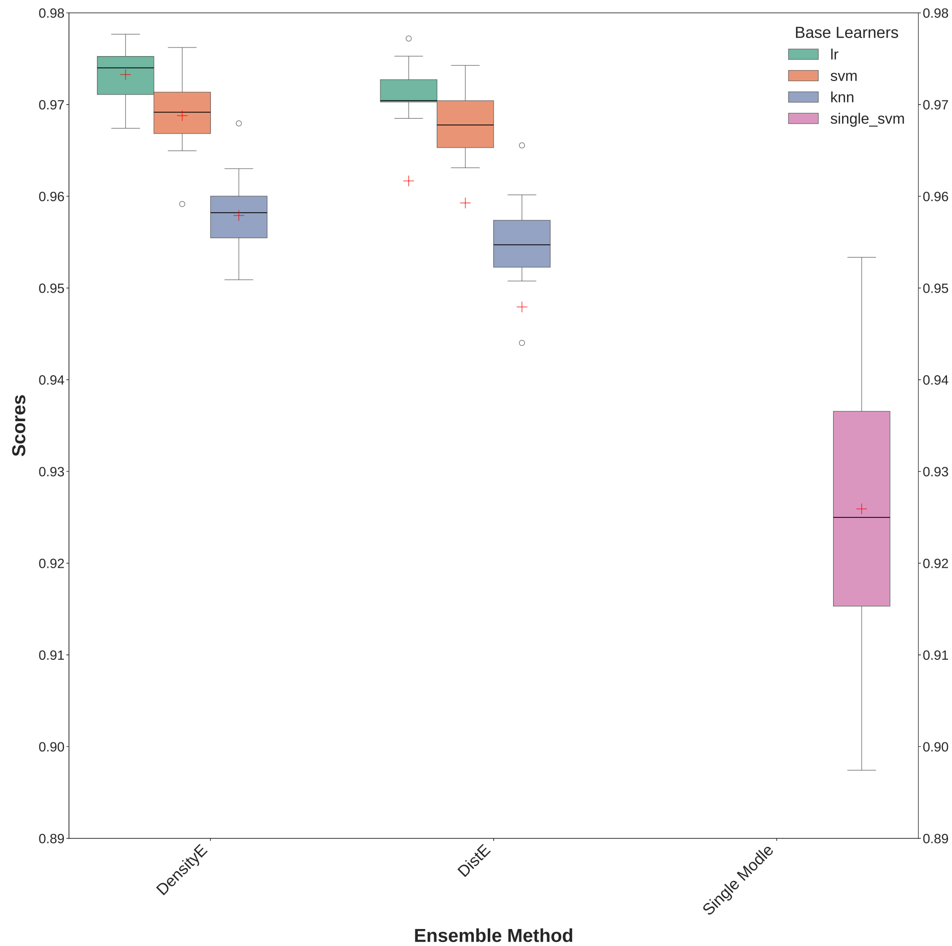

Fig. 5. Comparison of F-score Distributions Across Different Ensemble Methods and Base Learners

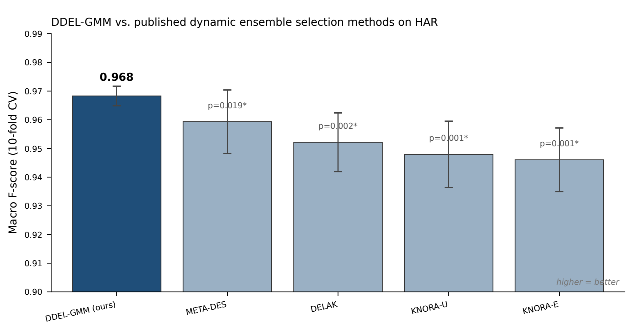

Fig. 6. Macro F-score of DDEL-GMM against published dynamic ensemble selection methods on the HAR dataset. Error bars are ± 1 standard deviation over the 10 cross-validation folds; annotations report the one-sided Wilcoxon signed-rank p-value of DDEL-GMM against each baseline ($^{*}p < 0.05$).

V. Discussion

The results unequivocally indicate that the DensityE ensemble method offers superior performance compared to other ensemble methods, particularly when logistic regression and support vector machine are utilized as base learners. This finding underscores the efficacy of DensityE's approach in leveraging the strengths of diverse base learners and effectively managing their combined output. The superior performance of logistic regression and support vector machine base learners can be attributed to their robustness and strong generalization capabilities, especially when applied to data with complex underlying distributions.

The comparative analysis across different base learners highlights the critical role of base learner selection in ensemble methods. While linear models like logistic regression and support vector machine benefit significantly from the ensemble approach, non-linear models such as k-nearest neighbors and decision trees show relatively lower performance. This disparity may be due to the inherent differences in how these models handle data complexity and variability.

The insights gained from Figures 1 and 2 provide a deeper understanding of ensemble dynamics. Increasing the number of clusters generally leads to better performance, but the benefits diminish beyond a certain point. Similarly, a higher sampling ratio enhances the F-score, emphasizing the value of larger training samples for achieving higher accuracy.

In conclusion, the DensityE ensemble method, particularly when paired with logistic regression and support vector machine base learners, demonstrates the most promising performance. Future research could explore the integration of more sophisticated base learners and the application of these ensemble methods to various types of datasets to further validate their robustness and generalizability. Additionally, investigating the effects of varying other parameters within the DensityE method may yield further performance improvements and insights.

VI. Conclusion

In this paper, we introduced the Density-Based Dynamic Ensemble Learning Algorithm using Gaussian Mixture Models (DDEL-GMM), a novel framework specifically engineered to enhance the personalization of predictions in complex data environments. The core challenge addressed is the limitation of traditional models in adapting to the unique characteristics of individual data points. DDEL-GMM tackles this by employing a sophisticated two-stage process. First, it uses Gaussian Mixture Models to partition the data into nuanced, overlapping clusters, capturing underlying data structures more effectively than methods based on rigid partitioning. This allows for the training of highly specialized base classifiers. Second, its density-based dynamic selection mechanism creates a bespoke, weighted ensemble for each individual test instance, ensuring that the final prediction is highly personalized and context-aware.

Our experimental evaluation confirmed the superior performance of this personalized approach. DDEL-GMM, particularly when paired with Logistic Regression and Support Vector Machine base learners, significantly outperformed other ensemble strategies. Beyond the internal aggregation variants, a direct comparison against published dynamic ensemble selection methods—DELAK, META-DES, and the KNORA family—showed that DDEL-GMM achieves the best macro F-score (0.968) and mean rank, with the improvements statistically significant under a paired Wilcoxon signed-rank test. Crucially, it addresses the key limitations of prior work such as the DELAK model. While DELAK utilizes K-means, which struggles with non-spherical data distributions, DDEL-GMM's use of GMMs allows for a more flexible and accurate modeling of complex data structures. The 1.6-point F-score margin over DELAK—which differs from our method chiefly in the choice of clustering algorithm—provides direct empirical support for this design decision. This leads to a more effective specialization of base learners and, ultimately, more precise predictions. The density-based selection further refines the model's competence assessment beyond simple distance metrics, enhancing its robustness and accuracy.

The findings establish DDEL-GMM as a powerful solution for classification tasks where personalization is paramount. The primary contribution of this work is a robust framework that advances the state-of-the-art in dynamic ensemble learning by delivering more accurate, reliable, and highly personalized predictive analytics.

Future research will aim to build upon this foundation. We intend to explore the integration of more diverse and complex base learners and to test the framework's generalizability across various domains where personalized modeling is critical, such as personalized medicine and financial risk assessment. Further investigation into optimizing the GMM clustering phase and adapting the framework for real-time data streams will also be key areas of focus.

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